

ASSIGNMENT (Aug 18, 2026)
RNA-Seq Literature Familiarization and Data Characterization

Preparation for RNA-Seq Re-analysis Using Galaxy and GitHub

Name	Balana, Jane Kate R.
Course and Section	BIO 300 - A
Date	August 18, 2026
Instructors	Ma'am Lilibeth Bucol & Sir Abner Bucol
Group Topics	Group 4 - Salinity Stress

Part 3. Understand the Paper Before Downloading Data

Information from the Paper	Group Answer
Title of the paper	The impact of salt stress on the physiology and the transcriptome of the model streptophyte green alga <i>Chara braunii</i> .
Authors	Daniel Heß, Carolin M. Heise, Hendrik Schubert, Wolfgang R. Hess, Martin Hagemann
Year	2023
Journal	Physiologia Plantarum, 175(6), e14123
Organism	<i>Chara braunii</i>
Tissue / organ / cell type	Subsurface thalli segments
Biological question	How does salt stress affect the physiology and gene expression of <i>Chara braunii</i> ?
Control condition	0 practical salt units (PSU)/ freshwater
Treatment or stress condition	5 PSU
Number of biological replicates	7 replicates
Sequencing platform	Illumina

Single-end or paired-end sequencing	paired-end sequenced
Read length, if reported	2x100 bp
RNA-seq repository	Sequence Read Archive
BioProject / Study accession	PRJNA965918
Run accession numbers	SRR24401886 - SRR24401893
Reference genome or transcriptome used by the authors	<i>Chara braunii</i> genome
Main RNA-seq software or tools reported by the authors	cutadapt, STAR, DESeq2, and goseq

Part 4. Prepare a Simple RNA-Seq Experimental Design Diagram

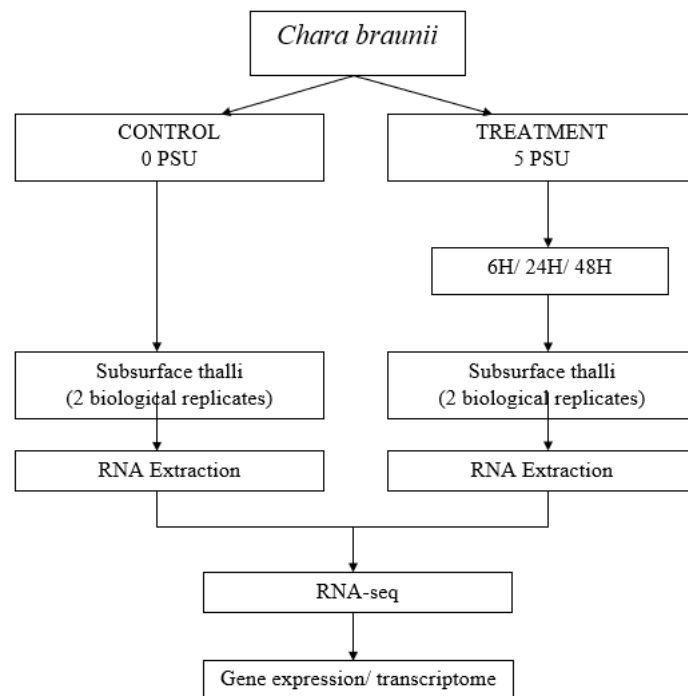


Figure 1. Experimental workflow for *Chara braunii* RNA-Seq analysis under salinity treatment.

- **What was changed or manipulated?**
 - The salinity level was manipulated by exposing *Chara braunii* to 5 practical salt units (PSU) salt stress for 6, 24, or 48 hours.
- **What was the control?**
 - The control was *C. braunii* maintained in freshwater at 0 PSU. The medium was replaced with fresh deionized water without added salt, providing the 0-hour/0-PSU reference condition.
- **What was measured?**
 - The researchers measured physiological responses and changes in gene expression (transcriptome) after salt stress. For the RNA-seq part, they measured gene expression/transcript levels.
- **How many biological replicates were used? What tissue or cell type was sequenced?**
 - For the RNA-seq experiment, 2 biological replicates (duplicates) were used for each sample. The sequenced material was subsurface thalli segments of *C. braunii*.

Part 5. Locate the Public RNA-Seq Data

SRX	Condition	SRR Run Accession	Control/Treatment	# of Spots	# of Bases	Approx. Size
SRX20189891	Salt stress t24_1f	SRR24401886	Treatment	72M	14.5G	4.2Gb
SRX20189890	Salt stress t6_2f	SRR24401887	Treatment	61.6M	12.4G	3.6Gb
SRX20189889	Salt stress t6_1f	SRR24401888	Treatment	56.3M	11.4G	3.3Gb
SRX20189888	Salt stress t0_2f	SRR24401889	Control	57.1M	11.5G	3.4Gb
SRX20189887	Salt stress t48_2f	SRR24401890	Treatment	48.9M	9.9G	2.9Gb

Part 6. Assign One RNA-Seq Sample to Each Student

Run Accession Number	Condition	Biological Replicate Number	Single-end or Paired-end	Approximate File Size
SRR24401886	Treatment	Rep 1	Paired-end	4.2 Gb

Part 7. Download the RNA-Seq Data into Galaxy

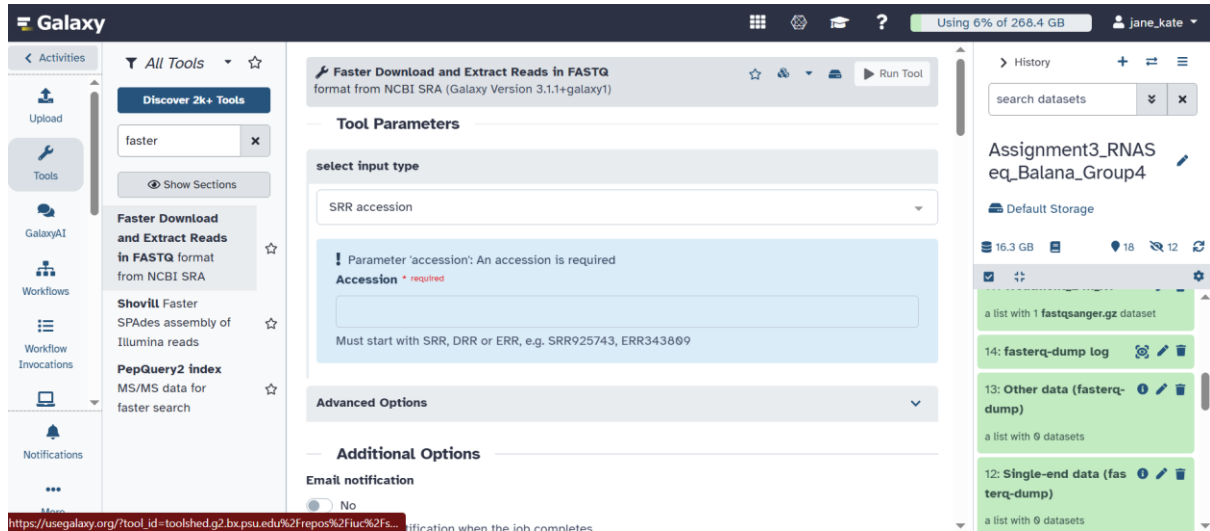


Figure 2. Download and extract of Pair-End and Single-End RNA Seq Reads in Galaxy using the tool FASTER Download and Extract Reads in FASTAQ.

Part 8. Examine the FASTQ File

Matrix	Control_Rep_R1	Control_Rep1_2
Line 1: sequence identifier	@SRR24401886.1/1	@SRR24401886.1/2
Line 2: nucleotide sequence	CAATCGTCCGCCACCGC TCGCTTTATAATCCTCC GCCACCGCCTCACGCAA ACACGCAGCTTTCCTT ACTCATCCCCCTTCGTG TTGTGACCACCTGAAA	ATCATCAAGTCATTGTG TCATACCTGGAAACCCA TCTCCTTCTGTCCCTGG TTAGTCGTGAGGACTTG GTTTTTCAGGTGGTCACA ACACGAAGGGGGATGA
Line 3: separator line	+	+
Line 4: quality-score characters	FFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFFFFFFFFFF FFFFFFFFFFFF:FFFFFFFF FFFFFFFFFFFFFFFFFFFF	FFFFFFFFF:FFF:F,FFFFFF FFFFFFFFFFFFFFFFFFFF,FF FFFFFFFF,FFFF,FFFFFF,F,F ,F:FF:,FFFF:FFFF,FFFFFF FFFFF,FFFFF

1. How is a FASTQ file different from the FASTA genome file used in the previous assignments?
 - A FASTQ file contains both nucleotide sequences and their corresponding sequencing quality scores, while a FASTA file primarily contains the sequence and its identifier. In the examined RNA-seq data, each read is represented by four lines: a sequence identifier, the nucleotide sequence, a separator line (+), and the quality-score characters.
2. What information is present in FASTQ that is not present in FASTA?
 - FASTQ contains quality-score information for each nucleotide base. These scores indicate the reliability of the bases generated during sequencing. This information is not included in a standard FASTA sequence file.
3. Why are quality scores important in RNA-seq?
 - Quality scores are important because they indicate the accuracy and reliability of the nucleotide sequences. Low-quality bases may contain sequencing errors that can affect read alignment and downstream analyses, such as transcript identification and gene-expression analysis. Therefore, quality scores help researchers assess the overall quality of RNA-seq data.



Figure 3. Preview of the forward reads from paired-end RNA sequencing data in FASTQ format, generated using the Galaxy tool Unzip Collection to Forward Datasets.



Figure 4. Preview of the reverse reads from paired-end RNA sequencing data in FASTQ format, generated using the Galaxy tool Unzip Collection to Reverse Datasets.

Part 9. Run FastQC

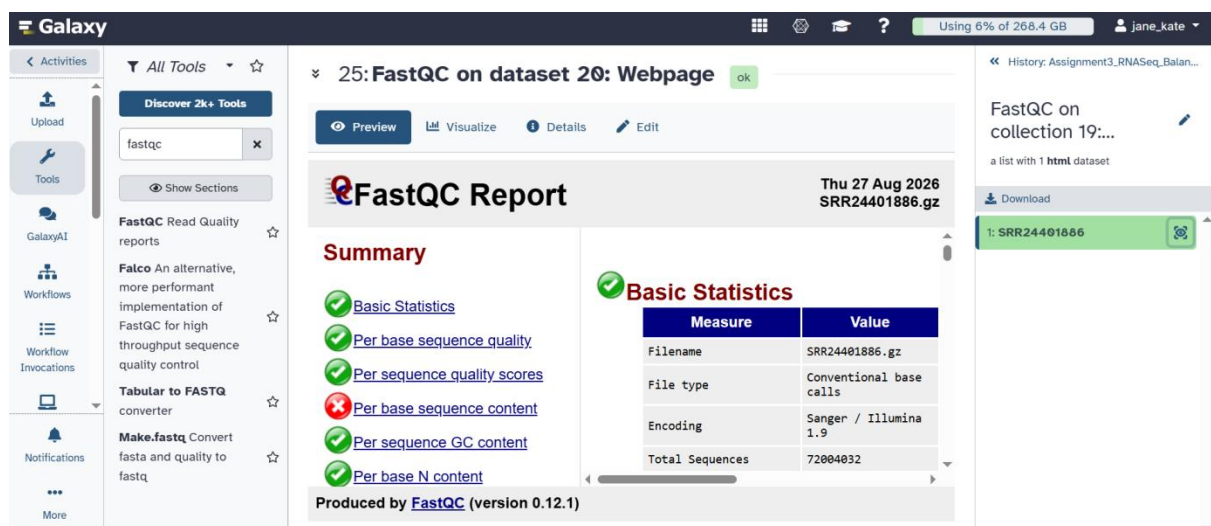


Figure 5. FastQC Quality Control for Forward Read generated using the FastQC tool in Galaxy.

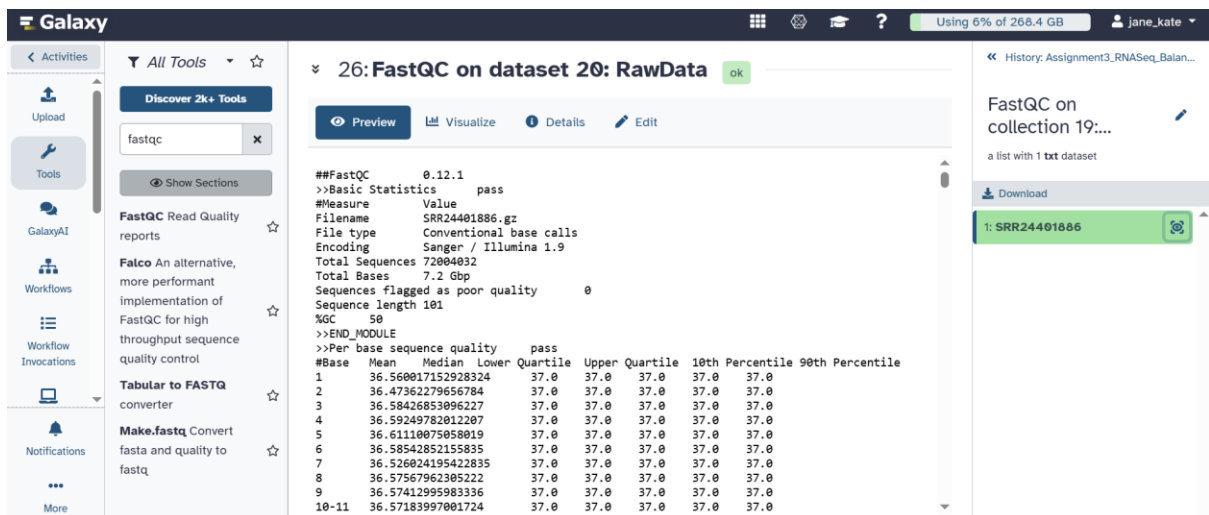


Figure 6. Raw Data Output of the FastQC Analysis for the Forward Read Generated Using the FastQC Tool in Galaxy



Figure 7. FastQC Quality Control for Reverse Read generated using the FastQC tool in Galaxy.

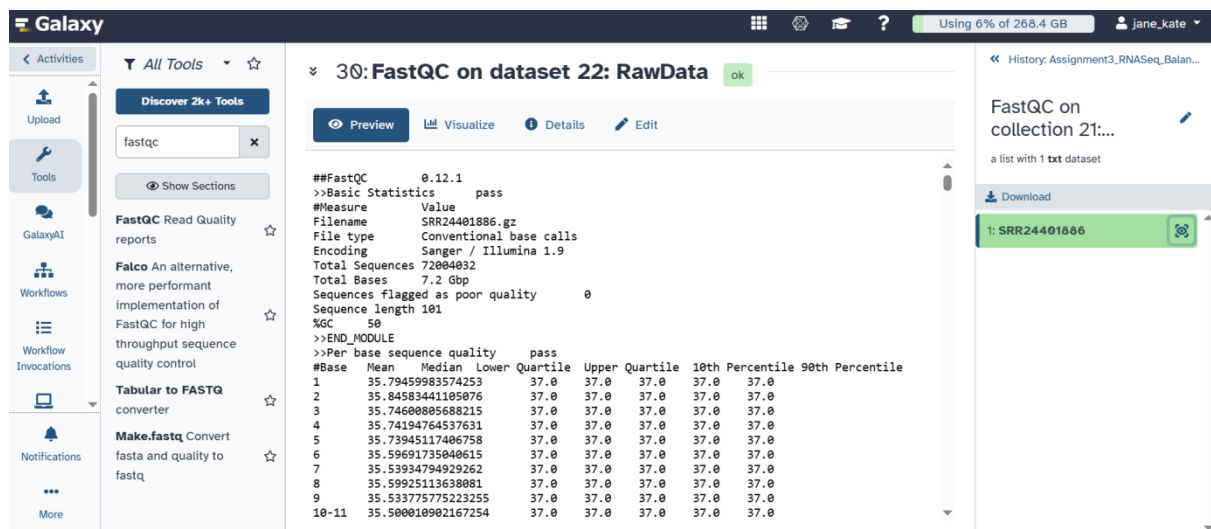


Figure 8. Raw Data Output of the FastQC Analysis for the Reverse Read Generated Using the FastQC Tool in Galaxy

Part 10. Characterize Your RNA-Seq Dataset

RNA-Seq Characteristics	Your Observations
RNA-seq run accession	SRR24401886
Condition	Treatment – salt stress, 24 h (t24), Rep1
Single-end or paired-end	Paired-end
Number of reads / sequences	R1: 72,004,032 R2: 72,004,032
Read length	R1: 101 bp R2: 101 bp
File size	R1: 2.9 GB R2: 2.9 GB
GC content	R1: 50% R2: 50%
General per-base sequence quality	R1: Pass R2: Pass

Presence of adapter content	R1: Pass R2: Pass
Presence of overrepresented sequences	R1: Warn R2: Fail
Any quality warning observed	R1: Per base sequence content (fail), Sequence Duplication Levels (fail) R2: Per sequence GC content (warn), Per base sequence content (fail), Sequence Duplication Levels (fail), Overrepresented sequences (fail)

Part 12. Compare the RNA-Seq Samples Within the Group

(As a group, combine the observations from all members. Prepare a small comparison table.)

Student	Accession	Condition	Reads	Read length	GC %	Main QC observation
Balana, Jane Kate R.	SRR2440188 6	Treatment – salt stress, 24 h (t24)	72,004,032	101 bp	50%	PASS – Good overall sequencing quality with a high number of reads, 101 bp per-mate read length, and a balanced GC content of 50%. No major quality issues are indicated.
	87					
Trayvilla, Shekhan Fredianne C.	SRR2440188 8	Treatment - salt stress, 24 h (t24)	56,332,223	101 bp	50%	PASS – Excellent overall sequencing quality with high per-base quality (mean Q ≈34–36), no significant N content, and consistent 101 bp reads. However, poly-G sequences, high

						duplication, and biased per-base sequence composition were detected.
Olaguir, Junavhel Jane	SRR24401889	Control - freshwater (t0), Rep2	57, 100, 000	101 bp	51%	PASS – Good overall quality. Per-base quality is high (Pass). No adapter contamination. Some warnings for sequence duplication and overrepresented sequences, which is common in RNA-Seq.
Catipay, Nina Karylle	SRR24401890	Treatment - salt stress, 48 h (t48)	48,881,896	101 bp	51%	WARNING – Adapter content detected (Warn). Sequence duplication levels failed. Per-base sequence content failed. This sample has more quality issues than others, likely due to library prep or high expression of specific genes.

Part 13. Interpretation Questions

- What biological question was the original RNA-seq study trying to answer?
 - The study wanted to determine how *Chara braunii* responds and acclimates to salt stress at the physiological and gene-expression levels. The researchers exposed the algae to 5 PSU salt stress for 6, 24, and 48 hours and used RNA-seq to identify changes in gene expression. They were particularly interested in which genes were activated or suppressed during salt acclimation and what biological processes were involved.
- Why did the authors use RNA-seq instead of only examining the genome?
 - The genome shows what genes an organism has, but it does not tell us which genes are actively being expressed under a particular condition. RNA-seq allowed the researchers to examine the transcripts (mRNA) being produced by *C. braunii* during salt stress. This helped them identify genes whose expression increased or decreased after exposure to 5 PSU salt. The study quantified transcript levels for 17,387 genes and identified genes involved in processes such as ion/solute transport and cell-wall modification.

3. What is the difference between genomic DNA and the RNA molecules measured by RNA-seq?

- Genomic DNA is the organism's relatively stable genetic information and contains the genes and regulatory regions that make up its genome. RNA molecules, particularly mRNA, are produced when genes are expressed. Therefore, genomic DNA tells us what genes are present, while RNA-seq measures which genes are being expressed and how much transcript is being produced at a particular time and condition.

In this study, the researchers extracted total RNA from *C. braunii* thalli, isolated the mRNA, converted it into cDNA libraries, and sequenced those molecules.

4. What is a biological replicate and why is it important?

- A biological replicate is an independent biological sample that represents the same experimental condition. It is important because biological organisms naturally vary, so using replicates helps researchers determine whether an observed difference is due to the treatment rather than random biological variation.

In this study, the RNA samples were prepared in biological duplicates for each sample, meaning each RNA-seq condition had two independent biological replicates. The researchers could then compare the replicates to assess whether their gene-expression responses were consistent.

5. What is the difference between single-end and paired-end sequencing?

- In single-end sequencing, each RNA-seq fragment is sequenced from one end, producing one sequence read per fragment.

In paired-end sequencing, each fragment is sequenced from both ends, producing two reads: R1 (forward) and R2 (reverse). These two reads correspond to the same original fragment and provide more information about its sequence and location.

6. What is a FASTQ file?

- A FASTQ file is a text-based format used in bioinformatics for storing both a biological sequence and its corresponding base quality scores. Each sequence entry typically consists of four lines: a sequence identifier header starting with @, the raw sequence string, a separator header starting with +, and quality score characters for each base.

7. What information does FastQC provide?

- FastQC is a quality control tool that analyzes raw high-throughput sequencing data and generates a summary report. It provides metrics such as, per-base and per-sequence quality scores, per-base sequence content, per-sequence GC content distribution, per-base N content, sequence length distribution, sequence duplication levels and overrepresented sequences, and possible adapter content.

8. What does a high per-base quality score indicate?
 - A high per-base quality score indicates high confidence in the accuracy of the base call. It means the probability that a base was called incorrectly by the sequencing instrument is extremely low.
9. Why can adapter contamination be a problem?
 - Adapter contamination happens when sequencing reads accidentally include artificial laboratory adapters. Because these extra sequences do not belong to the organism, they confuse computers during analysis, leading to mapping errors and incorrect gene measurement. Trimming these synthetic pieces out ensures the final data is accurate and clean.
10. Were all RNA-seq samples in your group similar in quality? Explain.
 - Mostly yes, but not entirely. Most samples passed the per-base quality and adapter checks. However, some samples (like SRR24401890) showed warnings for adapter content and sequence duplication, while others had more severe failures in duplication levels. This suggests some variation in library preparation or biological complexity.
11. Did any sample show a possible quality problem? What was it?
 - Yes. SRR24401890 (Catipay's sample) showed a "Warn" for Adapter Content and "Fail" for Sequence Duplication Levels. SRR24401891 (Itom's sample) also had multiple "Fail" warnings for duplication and overrepresented sequences. These issues may require trimming or filtering before analysis.
12. What additional steps would be needed before the researchers could compare gene expression between control and treatment samples?
 - To compare gene expression between control and treatment samples, researchers must first trim adapter sequences and filter out low-quality bases using Cutadapt. Next, the cleaned reads are aligned to the *Chara braunii* reference genome using STAR. The mapped reads are then quantified for each gene to build a raw expression count matrix. Finally, this data is normalized and statistically analyzed using DESeq2 to identify significantly differentially expressed genes.

Part 15. Summary

This activity evaluated the RNA-Seq data from *Chara braunii* under salt stress (5 PSU) across multiple exposure times using tools in Galaxy. The quality control assessment showed that while most samples exhibited high per-base sequence quality and consistent GC content, several datasets (such as SRR24401890) flagged warnings or failures due to sequence duplication, adapter content, and overrepresented sequences. These findings highlight the importance of pre-processing steps, such as adapter trimming and duplicate removal, prior to downstream analysis. Ultimately, quality-checked raw reads are essential to ensure accurate alignment and reliable differential gene expression profiling during salt acclimation.

Article Link:

Heß, D., Heise, C.M., Schubert, H., Hess, W.R. & Hagemann, M. (2023) *The impact of salt stress on the physiology and the transcriptome of the model streptophyte green alga Chara braunii*. *Physiologia Plantarum*, 175(6), e14123. Available from: <https://doi.org/10.1111/ppl.14123>